

- (c) Sodium fluoride is added to some UK public water supplies to reduce tooth decay in children.

In America sodium hexafluorosilicate,  $\text{Na}_2\text{SiF}_6$ , is more commonly used. The relative formula mass of sodium hexafluorosilicate is 188.

- (i) Calculate the percentage of fluorine in sodium hexafluorosilicate. [2]

$$A_r(\text{F}) = 19 \qquad M_r(\text{Na}_2\text{SiF}_6) = 188$$

Percentage = ..... %

- (ii) State an **ethical** reason why some people oppose the fluoridation of water supplies. [1]

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- (iii) Apart from water supplies, state the most commonly used source of fluoride to reduce tooth decay. [1]

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